

Reference Sheet for Different Options

Vanilla Options

Option	Payoff
European Call Can only be exercised at expiration.	$C = \max(S_T - K, 0)$
European Put Can only be exercised at expiration.	$P = \max(K - S_T, 0)$
American Call / Put Same as European but can be exercised any time before expiration.	Same as European option payoffs, but with early exercise possibility.

Spread and Combination Strategies

Strategy	Payoff
Bull Call Spread Buy call at K_1 , sell call at $K_2 > K_1$.	$\min(\max(S_T - K_1, 0), K_2 - K_1)$
Straddle Buy one call and one put at same strike K .	$\max(S_T - K, 0) + \max(K - S_T, 0)$
Strangle Buy OTM call and put with strikes $K_1 < K_2$.	$\max(S_T - K_2, 0) + \max(K_1 - S_T, 0)$

Exotic Options

Option	Payoff
Asian Arithmetic Average Call Payoff based on average underlying price over time.	$\max\left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0\right)$
Asian Geometric Average Call Uses geometric average.	$\max\left(\left(\prod_{i=1}^n S_{t_i}\right)^{1/n} - K, 0\right)$
Lookback Floating Strike Call Depends on minimum price during option life.	$\max(S_T - \min_{t \in [0, T]} S_t, 0)$
Lookback Floating Strike Put Depends on maximum price during option life.	$\max(\max_{t \in [0, T]} S_t - S_T, 0)$
Up-and-Out Call (Barrier) Expires worthless if barrier breached.	$\begin{cases} \max(S_T - K, 0) & \text{if } \max_t S_t < B \\ 0 & \text{otherwise} \end{cases}$
Digital (Binary) Call All-or-nothing payoff if $S_T > K$.	$Q \cdot \mathbb{1}_{\{S_T > K\}}$
Chooser Option Choose between call or put at future time t_c .	$\max(S_T - K, K - S_T)$
Compound Option Option on another option.	$\max(\text{Call}(S_T, K_2) - K_1, 0)$
Passport Option Profit from active self-financing trading strategy.	$\max(\Pi_T, 0), \quad \text{where } d\Pi_t = \Delta_t dS_t$